MATEST

BE ALWAYS IN SAFE HANDS

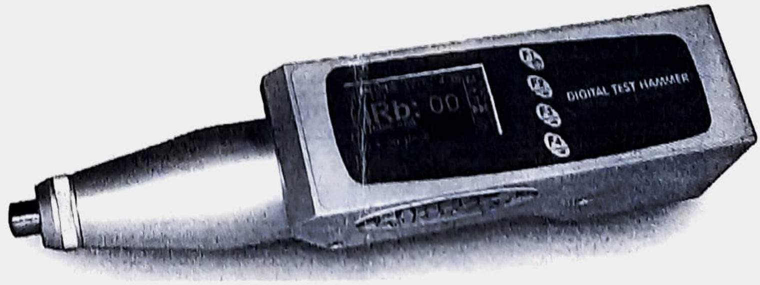
INSTRUCTION MANUAL

DIGITAL CONCRETE REBOUND HAMMER

MANUAL CODE

C386N

Do not attempt to operate this equipment before reading and comprehending the manual in all its parts



USERS: MACHINE MANUFACTURERS | DRAUGHTSMEN |OPERATORS | MAINTENANCE WORKERS | ANY OTHERS

REV.	DESCRIZIONE	REDATTO/GESTITO	APPROVATO	COD. IDENT.	PAGINE	DATA EMIS.
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ENCLOSURE | DRAWING AND EXPLODED VIEW

Chapter 1 GENERAL INFORMATION

1.01 WARNINGS

The manufacturer does not accept any responsibility for direct or indirect damage to people, things or animals and use of the appliance in different conditions from those foreseen.
The manufacturer reserves the right to make changes to the documentary information or to the appliance without advance notice.
Check the machine responds to the standards in force in the state in which it has been installed.
All operations necessary for maintaining machine efficiency before and throughout use are the operator's responsibility
Carefully read the entire manual before operating the machine.
It is vital to know the information and limitations contained in this manual for correct machine use by the operator.
Interventions are only permitted if the operator is accordingly competent and trained.
The operator must be knowledgeable about machine operations and mechanisms.
The purchaser must ensure that operators are trained and aware of all the information and clarifications in the supplied documentation.
Even with such certainty the operator or user must be informed and therefore aware of potential risks when operating the machine.
Safety, reliability and optimum performance is guaranteed when using original parts.
Any tampering or modifying of the appliance (electrical, mechanical or other) which has not been previously authorised in writing by the manufacturer is considered abusive and disclaims the constructor from any responsibility for any resulting damage.
All necessary operations to maintain the efficiency of the machine before and throughout use are the responsibility of the user

1.02 WARNING AND DANGER INDICATIONS - SIGNS

The machine has been designed and constructed according to the current norms and consequently with mechanical and electrical safety devices designed to protect the operator or user from possible physical damage. Residual risks during use or in some intervention procedures on the device are however present. Such risks can be reduced by carefully following manual procedures, using the suggested individual protection devices and respecting the legal and safety norms in force.
This manual includes "Warning" and "Danger" indications in relevant chapters. These indications are shown with the words "Danger" or "Warning" in bold font and uppercase to make them highly visible.

!	**"WARNING"**	Indicates that machine damage could be caused should indications be ignored.
⚠	**"DANGER"**	Indicates that machine damage and/or injury to the worker could be caused should indications be ignored.

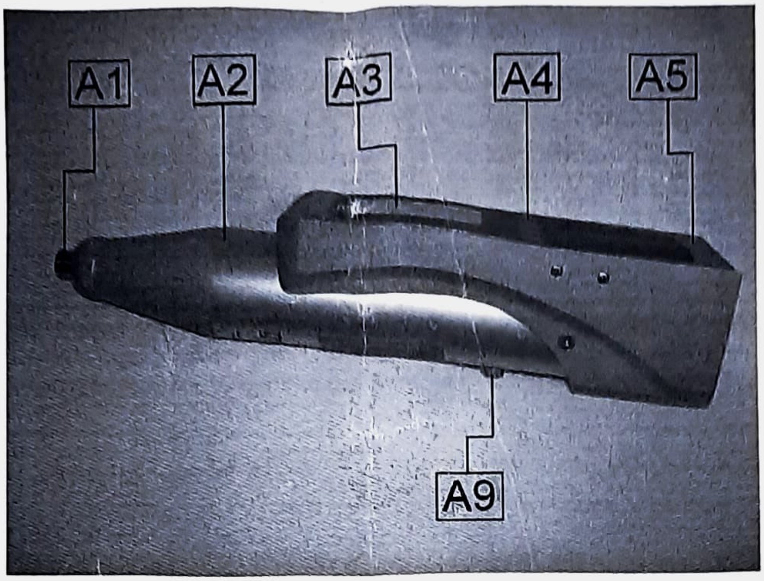
"DANGEROUS ZONE" indicates any zone inside or in the proximity of the appliance in which a person is exposed to the risk of injury or damage to health.

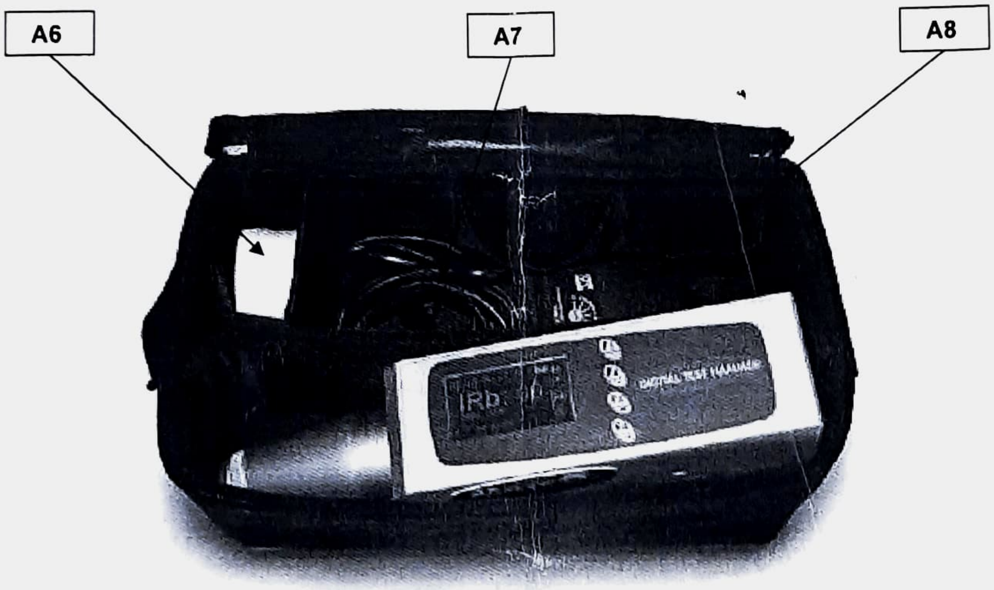
1.03 AIM OF THE INSTRUCTIONS MANUAL

This manual has been edited with the aim of providing all machine operators with all the necessary information on installation, use and maintenance from production to scrapping in as comprehensive and clear manner as possible.
All the procedures useful for any foreseeable emergency situations have been listed by the manufacturer and can be verified during use.
Operators, for whom this manual has been written, due to their competence must give instructions or operate the machine themselves.
The instructions manual must be carefully consulted by laboratory or site safety managers, equipment operators and any internal and external maintenance workers.
The manual is integral to the product and refers to this appliance only.

Chapter 2 TECHNICAL CHARACTERISTICS

2.01 GENERAL MACHINE DESCRIPTION





	CONTROL	DESCRIPTION
A1	PERCUSSION STEM	Mass that give the "hit" to the concrete surface to test, just press to release the mass.
A2	MAIN BODY	Same at the analog version.
A3	DISPLAY	New display for a rapid visualization of the value.

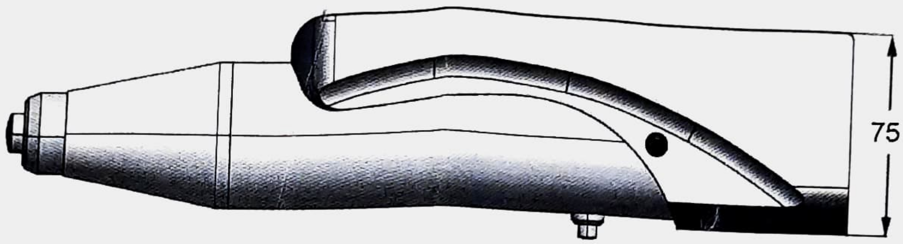
A4	FUNCTION BUTTONS	4 buttons allow to scroll between measures and menu.
A5	BATTERY CONTAINER	6 rechargeable battery type AA NIMh 2000mA/h
A6	ABRASIVE STONE	To smooth the surface of concrete to test.
A7	USB CABLE	Connect the instrument to the pc to download data
A8	BATTERY CHARGER	To recharge the instrument
A9	STOP BUTTON	It allows to block the percussion bracket inside the body of the instrument

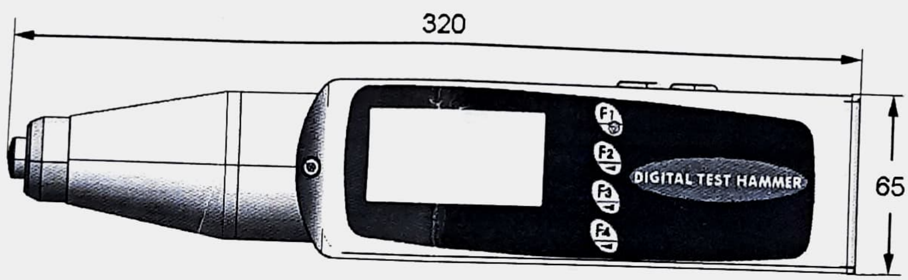
2.02 TECHNICAL SPECIFICATIONS

IMPACT ENERGY (CONCRETE VERSION)	2,207 Joule
IMPACT ENERGY (ROCK VERSION)	0,735 Joule

2.03 DIMENSION AND WEIGHT

LENGTH	320 mm
WIDTH	65 mm
HEIGHT	75 mm
WEIGHT	2 kg





2.04 ELECTRICAL SUPPLY

POWER UNIT/ BATTERY CHARGER	INPUT → 100 ÷ 240 VAC, 50 ÷ 60 Hz, 200 mA OUTPUT → 12 VDC, 700 mA (100 VAC), 800 mA (240 VAC)
BATTERIES	Ni-Mh 9 V, 2400 mA/h

The instrument functions with rechargeable batteries. The power unit supplied with the tool must be inserted to recharge the batteries.
The instrument does not have to be switched on for the battery to be recharged. The instrument can be switched on and used when being recharged.

N.B.: The batteries are usually completely recharged in 8 hours. Recharging stops automatically when the battery has reached its maximum charge.

Chapter 3 GENERAL SAFETY STANDARDS

3.01 GENERAL STANDARDS

To ensure the safety of machine operators:
· Any tampering with the appliance not pre-emptively authorised by the manufacturer exempts the manufacturer from any responsibility for damage caused by or to it.
· The removal or tampering with safety devices entails a violation of the safety standards.
· Machine use is only allowed in areas where there is no risk of explosions or fires.
· Only the original fittings can be used. The use of unoriginal fittings exonerates the manufacturer from all responsibility.
• Check the appliance is in ideal working conditions and that its parts are not worn or faulty before Carry out all necessary maintenance
• Be aware of the danger of electrical shocks from direct or indirect contact due to unforeseen electrical faults.
• Do not subject the appliance to violent impact.
• Do not expose the appliance to fire, welding sparks or extreme temperatures.
• Do not bring the appliance into contact with corrosive substances.
• Do not wash the appliance with jets of water.
· Check the workspace around the machine is clear from potentially dangerous objects.
· The machine operator must wear appropriate work clothing such as protective glasses, gloves and mask in order to avoid damage from, for example, harmful dust projection. Wear a lower back support when lifting heavy parts. There should be no hanging objects such as bracelets or otherwise, long hair should be protected with relevant precautions, shoes must be appropriate for the type of operation to be carried out.

DURING USE
When operating check there are no conditions of danger. Immediately stop the machine when it is functioning irregularly. Contact the authorised Sales Service department.

- • For the operator's safety do not touch any part of the appliance when testing and use the appropriate individual protection devices in order to keep the operator safe.

RISK OR DANGER	PROTECTION DEVICE
SQUEEZING OF THE HANDS OR OF THE ARMS	REINFORCED GLOVES
ABRASION - CUT	REINFORCED GLOVES

!	WARNING	**MAGNETIC FIELDS** Magnetic fields or the presence of strong intensity radiations must be avoided as far as possible. The static electricity or magnetic fields produced by other equipment may interfere with the internal components, resulting in damages to the data stored on the memory card or to the internal circuits of the instrument

!	WARNING	**CONTACT WITH LIQUID CRISTALS** In case of fortuitous breakdown of the display, pay particular attention not to injure you with the glass fragments and avoid that the liquid crystal could get in contact with the skin, eyes or mouth. The display is anyhow protected with an anti-shock protective film.

Tampering with the protections or any appliance modification could cause risks to users or other exposed people.
The manufacturer does not assume any responsibility for direct or in direct damage to people, things or animals following tampering with the protections.

3.02 MACHINE SAFETY DEVICES AND PROTECTION

DEFINITION: Protections are all the safety measures that consist of the use of specific technical means (repairs, safety devices) to protect people from dangers which cannot be limited reasonably in design.
The instrument does not have any safety devices as there are no particular risks for the operator during normal use.

Chapter 4 INSTALLATION INSTRUCTIONS

4.01 LOCATION

The equipment must be placed in an ideal position and environment for the use it has been conceived for (laboratory use and protected from atmospheric agents) and that the machine is placed by a qualified operator.

ALLOWED TEMPERATURE:	ALLOWED RELATIVE HUMIDITY:	MAXIMUM HEIGHT OVER SEA LEVEL:
from +5°C to +40°C	from 30% to 70%	1000 m

!	ATTENZIONE	**TEMPERATURE CHANGES** Avoid exposing the instrument to sudden change of temperature; sudden variation of temperature could generate condensation inside the body of the instrument with possible failures of the electrical components. We suggest to wrap the instrument or to put it back in its case in order not to shock it due to thermic change.

!	ATTENZIONE	**DO NOT WET THE INSTRUMENT.** The digital concrete Matest test hammer is not waterproof and may become faulty if placed where there is a high degree of humidity or immersed into water. The subsequent formation of rust in the components and internal mechanisms may cause irreparable damages.

4.02 ELECTRICAL CONNECTION

⚠	DANGER	Wiring of the electrical system must be carried out by qualified personnel Before wiring consult the electric plan linked to the instructions manual and the registration plate on the machine for information regarding supply, frequency and nominal current. Connect the earthing system via the PE terminal (yellow-green) before any other connection. Apply a knife switch at the top of the connecting cable of the machine to the power system. The knife switch must be combined with a safety device against the overload with a differential switch (safety switch). The technical features of the safety device must be in accordance with the standards in force in the country where the machine has been installed.

ELECTRIC TOLERANCES:

- Real voltage ± 10 % of the nominal one
- Frequency: ± 1 % of the nominal one in a continuous way
  ± 2 % of the nominal one for a short period
- The harmonic distortion of the sum from the second to the fifth harmonics not more than 10 % of the total voltage as a real value between the conductors. A further distortion of 2% is accepted for the sum from the sixth to the thirtieth harmonics of the real total value between the conductors.
- With reference to the voltage imbalance of the three-phase voltage, the inverted sequence component and the zero sequence component must not be more than 2% of the direct sequence component of the voltage.
- The voltage pulses must not last more than 1,5 ms with an up/down time between 500 ms and 500 ms and a peak value not higher than 200 % of the real value of the nominal tension.
- The electric supply must not be interrupted or zeroed for more than 3 ms at any time. Between two interruptions it must not take more than 1 s.
- The interruptions must not overcome 20 % of the tension peak for more than one cycle. Between two interruptions it must not take more than 1 s.

The manufacturer assumes no liability for any damages to people, things and animals caused by the non-compliance of the above instructions

Chapter 5 IN FUNCTION – USE

❗ ⚠	**DANGER WARNING**	Before setting the machine in motion it is essential that the Operator and Safety Manager have read the Instructions Manual and understood all parts of the machine and activities linked to it (Risks, Dangers, Functionality, Operation, Protections, Commands, etc.)

5.01 SELECTION AND PREPARATION OF THE TEST SURFACES

The concrete elements to be subjected to the test must be at least 100 mm in thickness and fixed inside a structure.
Smaller sample pieces may be subjected to testing as long as these pieces are rigidly supported.
Areas that contain the presences of gravel nests, flaking, course textures or others porous elements and in the proximity of significant inertia must be avoided.
It should also be avoided, by performing a preliminary rebar locator investigation, the carrying out of sclerometric strikes in areas of passing armatures and in the vicinity of pre-compression cables and wires.
In the selection of an area to be subjected to the test the following factors should be considered:

- Identification of the areas interested in the passage of armatures;
- Type of surface;
- Status of the surface humidity;
- Carbonatization;
- Movement of the concrete during the test;
- Evaluation of the damage level of the surface subject to the test;
- Test direction;
- Other appropriate factors as, for example, the type of concrete and the declare resistance class.

The area to be subjected to the test must be approximately 300 mm x 300 mm.
Ensure that the distance between the two points of impact are not less that 25 mm and that neither is less that 25mm from the edge.
The preparation of the test is carried out using an abrasive medium grain carborundum stone.
The smooth or float surfaces may be subjected to testing without rectifications.
Remove eventual water residue present on the cement surface.

5.02 SWITCHING ON THE EQUIPMENT

Before using the appliance it is highly recommended to check that the appliance and all its accessories work properly and are in good conditions, without defects or damaged parts

1. Pull concrete test hammer from the case box
2. Select a body with a high mass and a hard, smooth and flat surface
3. Grasp the body of the hammer firmly with both the hands out the
4. Lean the spherical edge of the percussion bracket **A1** on the selected surface
5. pushing the body of the instrument and keeping it perpendicular to the surface.
6. Push the appliance until the inner device that holds the percussion bracket will be unhooked.
7. Then gradually remove the instrument from the surface until the percussion bracket will be completely out. The obliquity of the impact plunger A1, towards the longitudinal axis of the instruments has to be considered normal and correct !
8. To start press the F1 button

The data on the main screen (seen when appliance is switched on) are as follows:

- Current date and hour

- Batteries status ( )or the presence of a mains supply ( ).
- Indication of the progressive number of the current series (**"TESTnnnn"**).
- The value of rebound of the last reading (**IRb**).
- The middle value of the rebound index (**IRbm**) referring to the current series of 10 readings.
- The value of Cubic Compression Strength (**CCS**), with the possibility to select as a unit (N/mm², psi, kg/cm²), ; the value is available in the range of CCS between 20 and 60, otherwise the advise "Na" (not available) will appear.
- The angle of rebound (α) referring to the current series of 10 readings (calculated upon first value reading and used to test the coherence of the following readings).
- The number of readings related to the current series (max 10). The symbology **N=nn/mm** indicates the display of the "nn" measure out of the "mm" totals (for example **N=01/04** indicates the first measure out of 4 totals).

5.03 PERFORMING TEST

1. Lean the spherical edge of the percussion bracket **A1** on the selected surface.
2. pushing the body of the instrument with both the hands and keeping it in a perpendicular position.
3. Apply a gradual pressure to the surface until the percussion is obtained.

!	ATTENTION	**DO NOT TOUCH THE SIDE STOP BUTTON "A9" WHILE YOU'RE PRESSING THE PERCUSSION STEM AND/OR WHEN THE HAMMER MASS IS MOVING, OTHERWISE THE INSTRUMENT WIL BE DAMAGED. THE STOP BUTTON MUST BE PRESSED ONLY TO MOVE THE HAMMER MASS BACK TO ITS HOUSING. ANY OTHER USE IS CONSIDERED INAPPROPRIATE.**

4. Keep the instrument firmly pressed against the surface and wait for the audio signal
5. Now is possible to remove the instrument from the surface The rebound index (IRb) shows the value of the latest test
6. Repeat the test maintaining the same tilt of the instrument (all the measurements must have coherent angling with the first one, otherwise data acquisition will not take place) up to a maximum of 10 readings per sequence with a wait of at least 2 seconds between one reading and the next.

If over 20% of all the measures fluctuates from the average by more than 6 units, the entire set of measurements taken will have to be discarded.

The value measured during the test and the current "job" are transferred to the USB (see chapter "TEST DESCRIPTION") so that they may be read and used by another instrument connected to it (i.e.: Ultrasonic tester).

Check all the tracks left by the instrument on the surface after the impact; if the impact has crashed or perforated the surface, do not consider the test result.
Humidity, carbonatization alterations, chemical aggressions, micro cracks, composition and history of the concrete, status of scabrous surface and underlying mass object of the percussions, are all elements that influence the bounce index value.
A correctly proportioned concrete presents a highly alkaline (pH13) environment which inhibits the oxidization reactions of the armature.
The concrete is however permeable therefore the carbon dioxide may distribute within reacting with the substances that it encounters giving way to the phenomenon of carbonatization (environment pH9) and to dimensional variations that determine the concrete cracks.
The cracking sustains the penetration of both carbon dioxide and water vapor which in turn triggers another process: the oxidization of the armature bars/rods, with notable effects.

How to save a sequence of measurements

1. Carry out at least one reading of the hammer value.
2. Press **F4** until appears.
3. Press to save the sequence.

How to cancel selection of a sequence of measurements
IMPORTANT! If the selected sequence has not been saved to file, the following procedure will cause the acquired data to be lost for the current sequence.

1. Check that a sequence of measurements has been selected (new or saved to file).

2. Press **F4** until S appears.
3. Press ⏭ to cancel selection of a sequence of measurements.

How to cancel a reading of a sequence of measurements

1. Carry out at least one hammer value reading.
2. Press ⟳ to scroll through the acquired measurements until the reading of interest is displayed.
3. Press **F4** until D appears.
4. Press ⏭ to cancel the sequence reading.

How to scroll through the readings of a sequence of measurements

1. Check a sequence of measurements has been selected (new or saved).
2. Press ⟳ to scroll through the saved measurements for the selected sequence: in the lower part of the screen the current reading will be displayed in real time and the total number of readings (e.g. **03/08**).

How to select a sequence of measurements from the file

IMPORTANT! If the selected sequence has not been saved to file, the following procedure will result in data loss.

1. Press M to access the selection menu.
2. Scroll down the menu by pressing + and − until is selected.
3. Press OK to access the file screen.
4. Press ↑↓ to select the test sequence to be selected (highlighted with "*").
5. Press OK to confirm (or ESC to cancel) the selection and return to the main menu.

How to delete a sequence of measurements from the file

IMPORTANT! If the selected sequence has not been saved to file, the following procedure will result in data loss.

1. Press M to access the selection menu.
2. Scroll down by pressing + and − until is selected.
3. Press OK to access the file screen.
4. Press ↑↓ to select the test sequence to be selected (highlighted with "*").
5. Press DEL followed by OK to delete the selected sequence: the selected measurement will be "marked" as "Cancelled" and will no longer be accessible.

How to delete the file

IMPORTANT! If the selected sequence has not been saved to file, the following procedure will result in data loss.

1. Press M to access the selection menu.
2. Scroll the menu by pressing + and − until is selected.
3. Press OK to access the file screen.
4. Press DEL followed by DEL to delete the entire file (the operation will take approximately 20 seconds) and return to the main screen.

5.04 TEST REPORT

The test report should include the following:

a) Identification of the element / concrete structure;
b) Position of the test area/s;
c) Test hammer identification;

d) Description of the test area/s preparation;
e) Concrete details and conditions;
f) Date and hour of the test run;
g) Test results (average value) and orientation of the test hammer for each test area;
h) Eventual deviations from the standardized test method;
i) Declaration of the person responsible for the test, whom can attest that the 12504-2:2001 test has been performed excepting that referred to in point (h).

Where necessary, the report may also include the single test hammer readings

5.05 CONFIGURATION

The instrument can be configured as follows:

1. Press M to access the selection menu.
2. Scroll down the menu by pressing + and − until  is selected.
3. Press OK to access the configuration screen.
4. Press ↑↓ to select the parameter to be modified (highlighted with "*") from:
   - Date and time.
   - Measuring unit to express the **CCS** value (N/mm², psi, kg/cm²).
   - Sample type undergoing test ("0" → concrete cylinder, "1" → concrete cube, "2" → rock cylinder, "3" → rock cube).
   - Correction coefficient of rebound index (multiplier applied to data read by instrument).
5. Press + and − to modify the value of the selected parameter.
6. Press ESC to stop modification, save the parameters and return to the main screen.

5.06 TEST DESCRIPTION

Some test descriptives can be defined using the appropriate screen. Proceed as follows:

1. Press M to access the selection menu.
2. Scroll down the menu by pressing + and − until  is selected.
3. Press OK to access the screen named "job".
4. Scroll through the descriptive parameters (in order: "JOB" - code "job", "nPIL" – number of pillars to be tested, "PIL" – pillar the test is referring to, "nFACE" – number of faces of pillars, - "FACE" face number the test is referring to, "ALPHA" – test angulation) which will "accompany" the data from the first sequence which will be filed by pressing + and −
5. Press OK to activate parameter modification.
6. Press + and − to modify the desired parameter.
7. Press ESC to deactivate parameter modification.
8. Press ESC to end modification and return to the main screen.

5.07 CALIBRATION

Proceed as follows for calibration:

1. Turn the instrument off.
2. Turn the instrument on keeping **F2**, **F3**, **and F4** pressed down. The calibration screen will appear with the following data:
   a. Theoretical rebound value for calibration anvil ("**Calibration Value**").
   b. Value of rebound index of last performed reading (**IRb**).

c. Median value of rebound index (**IRbm**) referring to the series of calibration.

3. Press + and − to modify the theoretical value of the rebound index for the calibration anvil used for calibration.
4. Perform series of 10 readings on the calibration anvil.
5. At the end of the last reading the main screen will be displayed.

IMPORTANT: The calibration procedure can be cancelled at any time (and return to the main screen) by pressing ESC.

Calibrating anvil characteristics
The verification of the calibration of an anvil does not guarantee that diverse test hammers will produce the same results in other points of the sclerometric scale.
In order to verify the calibration of the test hammer, the stainless steel anvil must be placed on a rigid surface.
Operate the instruments at least three times prior to initiating the readings from the calibration anvil, to ensure that the mechanics are operating correctly.
Then, following this produce, insert the test hammer in the anvil guide ring and carry out a series of strikes (no. = 10).

ATTENTION	The actual standards EN 12504-2 command the checking of the calibration of the test hammer with an anvil for each tests session (EN 12504-2 paragraph 6.1.3). The standard give mandatory indications on how verify and how often, before and after each test sessions. Measures taken on the reference anvil must be saved and compared (EN 12504-2 paragraph 6.3)

For this reason arise from the standard that is unnecessary a certificate of calibration for the test hammer, also if it is provided from the producer or another authority that use a different anvil of calibration, because the value obtained result not comparable with the ones obtained from the user before and after the test on his own anvil of calibration.

5.08 SWITCHING OFF

1. While in the main window, keep pressed F1 to turn off the instrument.
2. Lean the spherical edge of the percussion bracket **A1** on the selected surface.
3. pushing the body of the instrument with both the hands and keeping it in a perpendicular position.
4. Apply a gradual pressure to the surface until the percussion is obtained.

!	**ATTENTION**	**DO NOT TOUCH THE SIDE STOP BUTTON "A9" WHILE YOU'RE PRESSING THE PERCUSSION STEM AND/OR WHEN THE HAMMER MASS IS MOVING, OTHERWISE THE INSTRUMENT WIL BE DAMAGED. THE STOP BUTTON MUST BE PRESSED ONLY TO MOVE THE HAMMER MASS BACK TO ITS HOUSING. ANY OTHER USE IS CONSIDERED INAPPROPRIATE.**

5. Keep the instrument firmly pressed against the surface and wait for the audio signal
6. Keeping the instrument firmly pressed against the surface, press the **A9** stop button.
7. Gradually reduce pressure on the instrument and ensure the rod is kept inside.

5.09 BATTERIES

In order to grant a longer long-lasting of the batteries, the instrument switch off automatically the rear-lighting after 30 seconds when the machine is not in use.
After 15 minutes, the instrument switch off completely loosing the measurement not previously stored.

5.10 DATA TRANSFER SOFTWARE

ATTENTION	**The download software is normally compatible with the majority of the existing operating systems. Anyway considering: continuous evolution of these systems, several versions available on the market, potential programs configurations on PCs and eventual disturbs generated by some protection softwares like antivirus and firewalls, incompatibility cases cannot be excluded. Should an incompatibility occur or difficulties during installation or use be encountered, please contact Matest technical service.**

This procedure will allow you to install the software to transfer data from the instrument C386N to a standard personal computer. Be sure you have the enclosed CD Rom before starting.

1. Insert the CD Rom into the driver of the computer you want to download.
2. With the "autoplay function" ON, the installation procedure starts automatically; otherwise browse the CD and double click on the file **.<CD-ROM drive>:\setup.exe**.
3. Follow the screen instructions and confirming the default settings, install the software.
4. Double click on the file **<CD-ROM drive>:\Driver USB\OneSetup.exe** to start installation for the driver for USB connection.
5. Follow the screen instructions and confirming the default settings, install the driver.
6. Connect the USB cable between the instrument and the computer and wait for the periphery to be recognised.
7. Confirm with OK and NEXT on every window which appears until installation has been completed.

To transfer data from the rebound hammer, switch on the software, selection the correct communications port from the options menu and carry out data transfer of the instrument.

NB: In some O.S. it is mandatory to access with the account of Administrator to install the software.

How to send file data to the PC

IMPORTANT! If the selected sequence has not been saved to file, the following procedure could result in data loss.

1. Press M to access the selection menu.
2. Scroll down the menu by pressing + and − until  is selected.
3. Press OK to start sending data.
4. Install the driver for the serial communication:

    a. Double click on **<CD-ROM drive>:\Driver USB\OneSetup.exe** file, follow the instructions shown on the video, confirm the default settings, and wait for the end of the setup process.

    OR

    b. Double click on **<CD-ROM drive>:\Driver USB\CDM20814_Setup.exe** file and wait for the end of the setup process.

ATTENTION	**POWER CABLE AND DATA DOWNLOAD** Avoid detaching the instrument from the power supply or from the PC with the instrument switched on or during data transfer. Forced interruption of the power could cause data to be deleted or lost or damage to be caused to the internal circuit memory.

Chapter 6 MAINTENANCE

6.01 ORDINARY MAINTENANCE

Carry out a calibration check ("USAGE" CHAPTER) every time new tests are carried out.

DANGER WARNING	Do not perform maintenance – interventions on the machine which have not been quoted and described in this instructions manual without first contacting the manufacturer. Periodically clean all machine parts and oil the unpainted parts in order to preserve the machine and its efficiency. Avoid the use of solvents which damage paint and parts in synthetic material.

6.02 AUTHORISED MAINTENANCE CENTRES

For information on the nearest authorised help centre it is essential to contact the manufacturer.

Chapter 7 SPARE PARTS

WARNING DANGER	Only original spare parts can be used. Use of unoriginal spare parts exempts the manufacturer from all responsibility. Procedures for substitution of spare parts will be provided by the manufacturer along with the part. For spare parts contact the manufacturer's Sales Service department.

Chapter 8 INACTIVITY

Ensure all machine parts are in safe working order before operating it again should the machine be inactive for a long period of time. When in doubt contact the Manufacturer.

Chapter 9 DECOMMISSIONING THE MACHINE

Should it be decided that the machine is to be no longer used, proceed as follows:

- Disconnect the electrical supply network by removing the connecting cable therefore making it unusable.
- Make the potential sources of danger harmless, such as sharp or protruding parts.
- Dismantle the machine; divide it into similar parts and dispose of according to the standards in force.

Recycling notice for the disposal of electrical and electronical devices

This symbol, shown on the device or on the package and/or the documentation, suggests that the device should not be disposed together with other home garbage at the end of its life cycle.

To avoid further environment, or health-care damage, caused by the unsuitable disposal of garbage, the user should separate this device from other different types of garbage and recycle it in responsibly to avoid the reuse of material resources. Users must take care at the disposal of the equipment by taking it to the nearest recycling site for appropriate recycling treatment for electrical and electronical devices. Gathering and Recycling deplete devices allow the preservation of natural resources and grant them the adequate treatment by respecting health and environment.

For further information on your local recycling site please contact your local council or city waste treatment department. The developer, as producer of electrical and electronical devices, will provide to finance the recycling and treatment services for deplete devices that will come back through these recycling sites, according to the local statement.